Angular Advanced - Firebase – CRUD Operations

Peter Kassenaar –
info@kassenaar.com

ANGULAR

Firebase

CRUD operations in Firebase

- Options via AngularFire API – this is for `FirebaseListObservable` (!)
  - `push` – add new item
  - `update` – update item, use `$key` to indicate *which* item
  - `remove` – delete item. Also use `$key`.

API Summary

The table below highlights some of the common methods on the `FirebaseListObservable`.

method	
`push(value: any)`	Creates a new record on the list, using the Realtime Database's push-ids.
`update(keyRefOrSnap: string)`	Firebase
`remove(key: string?)`	Deletes the item by key. If no parameter is provided, the entire list will be deleted.

https://github.com/angular/angularfire2/blob/master/docs/3-retrieving-data-as-lists.md

Choice – add routing

- We add routing, so we can use Authentication and Guards later.
- Add `Routes, RouterModule` and routing table to application
- Add different component for different actions.
- Move logic and UI from `AppComponent` to `HomeComponent`.
- Add `<router-outlet>` and navigation menu to `AppComponent`.

```
<a class="btn btn-default" routerLink="home">
   <i class="fa fa-home"></i>
   Home
</a>
<a class="btn btn-default" routerLink="add">
   <i class="fa fa-plus"></i>
   Add City
</a>
<router-outlet></router-outlet>
```

Edit app.module.ts

```
const routes: Routes = [
   {path: '', pathMatch: 'full', redirectTo: 'home'},
   {path: 'home', component: HomeComponent},
   {path: 'add', component: AddComponent},
];

@NgModule({
   imports      : [
      …
      RouterModule.forRoot(routes)
   ],
   declarations: [
      AppComponent,
      HomeComponent,
      AddComponent
   ],
   providers    : [],
   bootstrap    : [AppComponent]
})
export class AppModule {
}
```

Add Routes

Add RouterModule

Add Components

Create CityService

- Create a Service, responsible for handling CRUD-operations.
- We *could* also create a Store, but leave that as an exercise to the reader ;-)
- Add `city.service.ts`, and CRUD-skeleton
- We also added a `City` interface as a Model

```
// city.model.ts
export interface City {
   id: number;
   name: string;
   province: string;
   highlights: string[],
   rating: number
}
```

CityService

```
@Injectable()
export class CityService {

   constructor(private af: AngularFireDatabase) {
   }

   getCities(): FirebaseListObservable<City[]> {
      return this.af.list('/cities');
   }

   addCity(city: City): void {
      //...
   }

   updateCity(city: City): void {
      //...
   }

   removeCity(city: City): void {
      //...
   }
}
```

TODO: fill this methods

HomeComponent

- Retrieve Cities via service

```
export class HomeComponent implements OnInit {
   cities$: FirebaseListObservable<City[]>;

   constructor(private cityService:CityService) {

   }

   ngOnInit() {
      this.cities$ = this.cityService.getCities();
   }
}
```

Inject CityService and set up Observable

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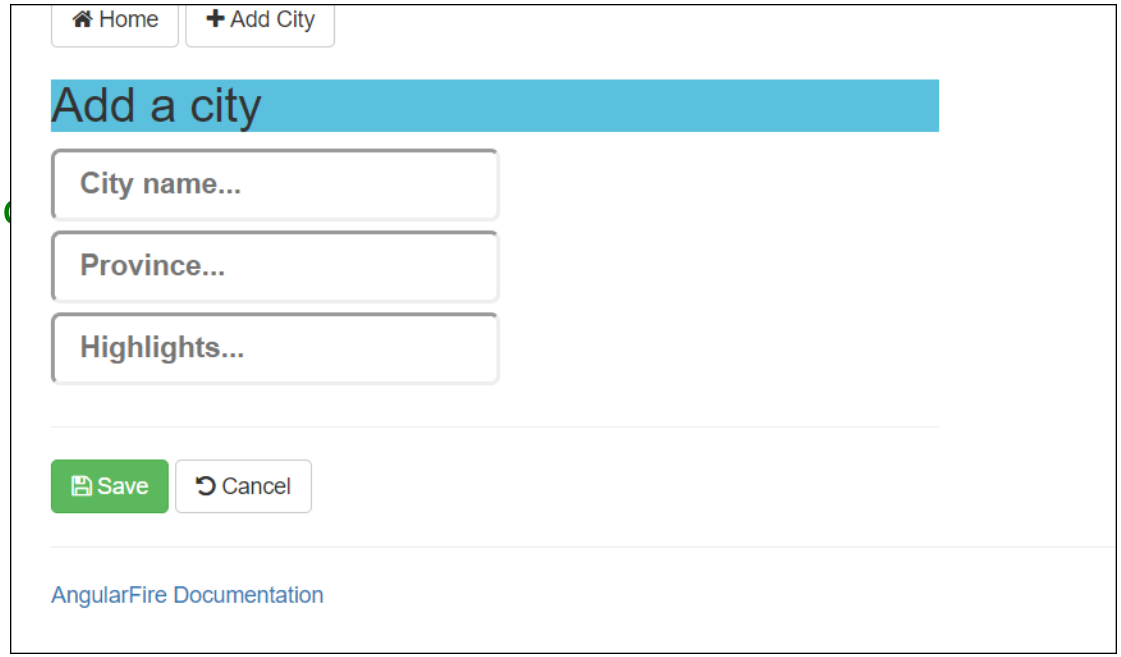
Adding a City

```
<!--Simple form to add a city. NO VALIDATION AT THIS MOMENT!-->
<input type="text" #cityName class="input-lg"
      placeholder="City name...">
<input type="text" #cityProvince class="input-lg"
      placeholder="Province...">
<input type="text" #cityHighlights class="input-lg"
      placeholder="Highlights...">
<button class="btn btn-success"
   (click)="addCity(cityName.value, c
   Save
</button>
```



Add logic

```
addCity(name: string, province: string, highlights: string): void {
   let newCity: City = {
      id        : this.cityService.getRandomId(),
      name      : name,
      province  : province,
      highlights: [],
      rating    : 0
   };
   this.cityService.addCity(newCity);
   this.router.navigate(['/']);
}
```

Component

```
addCity(city: City): any {
   return this.af.list('/cities')
      .push(city);
}
```

Service

Deleting cities

- Pass `$key` for unique reference
  - Remember to update your Model with property `$key`.

```
<li class="list-group-item"
   *ngFor="let city of cities$ | async ">
   {{ city.id}} - {{ city.name }}
   <button (click)="remove(city)" class="btn btn-sm btn-danger pull-right">
      <i class="fa fa-remove"></i>
   </button>
</li>
```

HTML

```
removeCity(city: City): void {
   this.af.list('/cities/' + city.$key)
      .remove();
}
```

Service

Home | Add City

List of Cities from Firebase

1 - Groningen

2 - Hengelo

3 - Den Haag

4 - Enschede

5 - Heerlen

AngularFire Documentation

Home | Add City

Add a city

Amsterdam

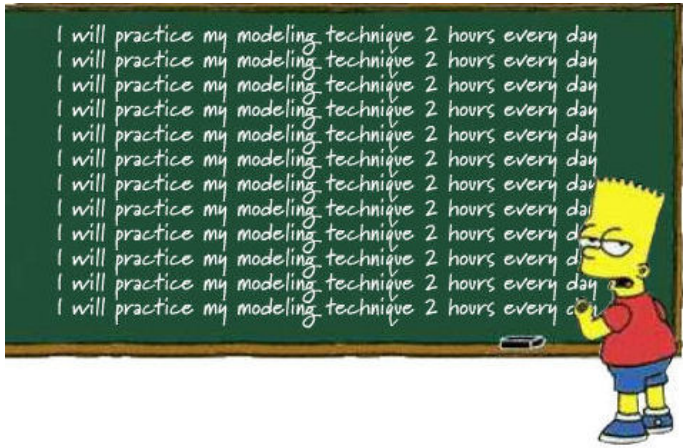
NH

Rijksmuseum

Save | Cancel

AngularFire Documentation

Workshop

- Add Routing to your project, create a `/home` route and `/add` route
- Add Component to add a new city
- Add Service and logic to Add and Delete cities in firebase

- Example: `/510-Firebase-crud`